Week 09: Limits of Machines: Gödelian Arguments and Dreyfus's Critique

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Two very different arguments claim that minds can do something that machines cannot. One appeals to Gödel's incompleteness theorems, the other to the phenomenology of skilled human activity. This week reconstructs both, examines the replies and asks which limits of machines are real [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to state the incompleteness theorem and the undecidability of the halting problem informally, and to construct a diagonal argument. They should be able to reconstruct the Lucas-Penrose argument and the consistency objection. Finally, they should be able to explain Dreyfus's critique of rule-based AI and its relation to later developments.

Lecture

Incompleteness and undecidability

Gödel proved in 1931 that any consistent formal system that can express elementary arithmetic contains statements that are true but cannot be proved within the system [1]. The construction uses self-reference: a sentence that, in effect, says of itself that it is not provable. If the system is consistent, the sentence is unprovable and therefore true. Turing reached a related result for machines [4]. No Turing machine can decide, for every program and input, whether the program halts. The proof is a diagonal argument: Given any claimed decider, one can build a program that asks the decider about itself and does the opposite. Figure 9.1 shows the idea as a table in which the flipped diagonal differs from every row.

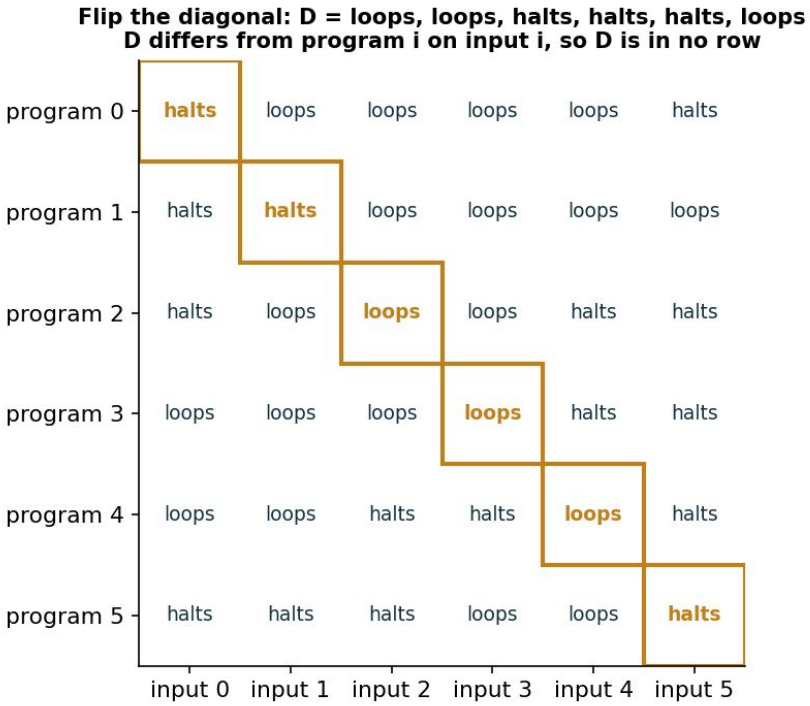


Figure 9.1. A diagonal argument: flipping the diagonal of any table of programs and inputs yields a behaviour that differs from every program in the table [4].

The Lucas-Penrose argument

Lucas argued that the incompleteness theorem shows that minds are not machines [2]. For any machine that proves arithmetic, there is a Gödel sentence that the machine cannot prove, but a human mathematician can see that the sentence is true. So the mathematician can do something that the machine cannot, and no machine can be an adequate model of the mind. Penrose developed a version of the argument and concluded that human mathematical insight is not computable, which led him to speculate about new physics in the brain [5, 6].

The central objection concerns consistency. A human sees that the Gödel sentence of a machine is true only if the human knows that the machine is consistent. Benacerraf argued that the argument establishes at most a disjunction: Either the mind is not a machine, or it is a machine whose program it cannot know or whose consistency it cannot prove [7]. Since people are not known to be consistent, and cannot prove their own consistency, the argument does not reach its conclusion. Most logicians and philosophers regard the Gödelian argument as unsuccessful, though it continues to be refined and debated.

Dreyfus's phenomenological critique

Dreyfus argued that the programme of symbolic AI rested on assumptions that were false [3, 8]. It assumed that the brain processes information in discrete operations, that the mind follows rules over symbols, that all knowledge can be formalised and that the world consists of independent facts. Drawing on Heidegger and Merleau-Ponty, he argued that human expertise is not rule-following. Experts respond directly to situations that show up as already meaningful

against a background of practices, and this know-how cannot be captured in explicit rules. The frame problem of Week 8 was, for Dreyfus, a symptom of this mistake.

Later, Dreyfus examined attempts to build Heideggerian AI, including behaviour-based robotics, and argued that they had not gone far enough [9]. In his view, Brooks's robots responded to fixed features of the world rather than to changing significance. A truly Heideggerian AI would need a body with needs that makes situations matter to it.

Which limits are real?

The two arguments differ in kind. The Gödelian argument claims an in-principle limit on any machine. It fails, according to most critics, because the same limits apply to humans. Dreyfus's critique targeted a particular approach, symbolic AI. Much of it has been vindicated by the shift to learning systems that acquire skills from experience rather than from explicit rules. What remains open is whether learned systems escape his deeper point about significance and embodiment. The undecidability of halting remains a genuine limit, but it binds every computing system, including brains, if they compute.

> ***Pause and reflect.*** *The interactive lab shows that a bounded halting checker is always wrong about some programs. Are you, as a human reasoner, subject to the same limit?*

Hands-on components

Interactive lab: Part A asks you to build the program D by flipping the diagonal of a table and to confirm that D appears in no row. Part B runs a halting checker that simulates each program for k steps. Part C tests Lucas's argument [2, 4, 7]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-09/lab.html

Colab notebook: It builds a table of programs and inputs from simple iterated maps, constructs the diagonal program, evaluates bounded halting checkers and detects cycles exactly where possible [1, 4]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-09/NB09_limits_of_machines.ipynb

Reflection prompts

1. Does undecidability limit human reasoners as well as machines? Explain how you would argue either way.
2. Which parts of Dreyfus's critique were vindicated by machine learning, and which remain challenges for present systems?
3. Choose a skill you possess that you could not reduce to rules. What does that suggest about the systems you build?

Weekly task and submission

What to do. Reconstruct and assess the Lucas-Penrose argument. First, write the argument as numbered premises and a conclusion. Second, explain the diagonal construction with your answer to Exercise 1 of the Colab notebook. Third, present the consistency objection. Finally, assess whether any version of the argument survives.

Where to look. Lecture: sections "Incompleteness and undecidability" and "The Lucas-Penrose argument". Interactive lab: Part A for the diagonal and Part C for Lucas's argument. Colab notebook: Section 1 with Exercise 1, which builds the diagonal program D, and Section 2 with Exercise 2 on bounded halting checkers.

Example. Lucas reasons in a way that resembles the diagonal construction of Exercise 1: For any machine, a human can see that some sentence is true although the machine cannot prove it. The consistency objection asks how the human knows that the machine is consistent, which this step requires.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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[8] Dreyfus, H. L. (1972). What Computers Can't Do: A Critique of Artificial Reason. Harper & Row.

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